The pigeonhole principle

High school · olympiad thinking · Lesson and method · 02

Prerequisites. Euclidean division and elementary counting.

Key point · Ideas and method. If $n+1$ objects are placed in n boxes, at least one box contains two objects. Otherwise, each box would contain at most one, giving at most n objects.
More generally, if N objects occupy k boxes, one box contains at least $\lceil \frac{N}{k} \rceil$. The ceiling $\lceil x \rceil$ is the smallest integer greater than or equal to x.
Choosing the boxes is the key: remainders, birth months, small grid squares or classes of integers. Explain why every object belongs to one box.

Example 1. Among 13 people, two were born in the same month: the 12 months are the boxes. This does not guarantee two birthdays on the same day.

Activity 1 · Explore. Choose six distinct integers from 1 to 10. Group them into pairs summing to 11 and find a complete pair.